Chapter 16, Problem 6RQ

Problem

If the input to a linear system is $\delta(t)$ and the output is $e{-}2tu(t)$, the transfer function of the system is:

(a) $\frac{1}{s+2}$

(b) $\frac{1}{s-2}$

(c) $\frac{s}{s+2}$

(d) $\frac{s}{s-2}$

(e) None of the above

Step-by-step solution

Step 1 of 1

The Transfer function of the system

$$=\frac{1}{s+2} \qquad \left[\therefore\ L\left[e^{-at}\right]=\frac{1}{s+a}\right]$$

$\boxed{a}$